problem 12.8

$60^{o}C$

$$\Delta Q_{b} = \frac{1}{2} \times 20 \times 10^{-3} \times 400^{2}\,\mathrm{J} = mc(T_{f} - 333\mathrm{K})$$

$$T_{f} = 333\mathrm{K} + \frac{\frac{1}{2} \times 20 \times 10^{-3} \times 400^{2}}{20 \times 10^{-3} \times 400}\mathrm{K} = 533\mathrm{K}$$

It is an irreversible process. Find a reversible process to connect same pair of states. In that case we can calculate as

$$\Delta S = \left(\int_{60+273}^{T_\mathrm{f}} \qquad + \int_{T_\mathrm{f}}^{15+273} \qquad \right) \frac{mc\mathrm{d}T}{T} + \frac{\Delta Q_s}{T_s}$$

Large barrier

$$= 20 \times 10^{-3} \times 400 \times \ln \frac{273+15}{273+60} \mathrm{J/K}$$

$$+ \frac{\frac{1}{2} \times 20 \times 10^{-3} \times 400^2 + 20 \times 10^{-3} \times 400 \times 45}{273+15} \mathrm{J/K}$$

$$= -1.16\mathrm{J/K} + 6.81\mathrm{J/K} = 5.65\mathrm{J/K}$$

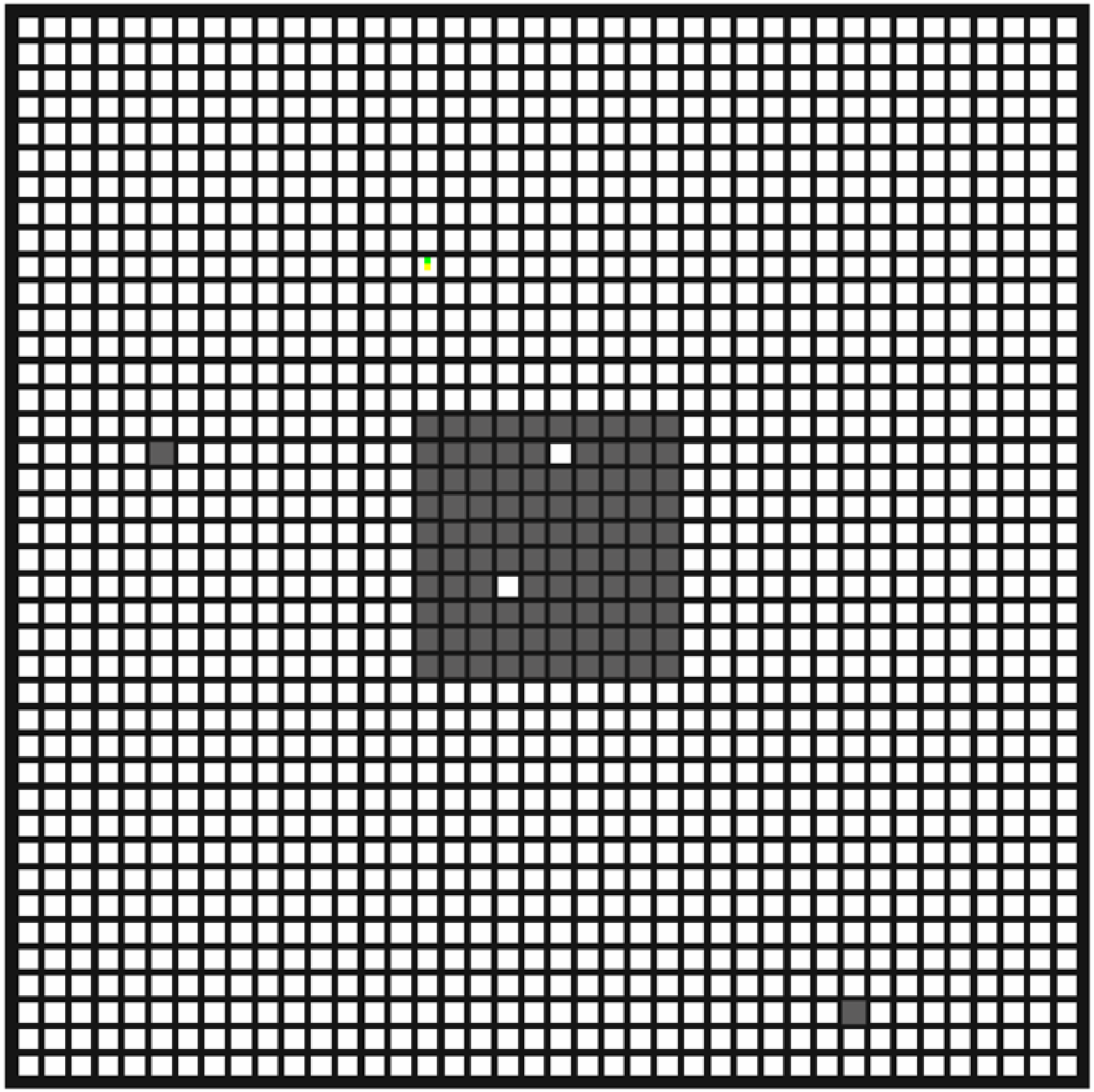
problem 12.8

Find a reversible process to replace the process that the bullet release heat to barrier.

Large barrier

$$\Delta S = \int_{60+273}^{15+273} \frac{mc\mathrm{d}T}{T} + \frac{\Delta Q_s}{T_s}$$

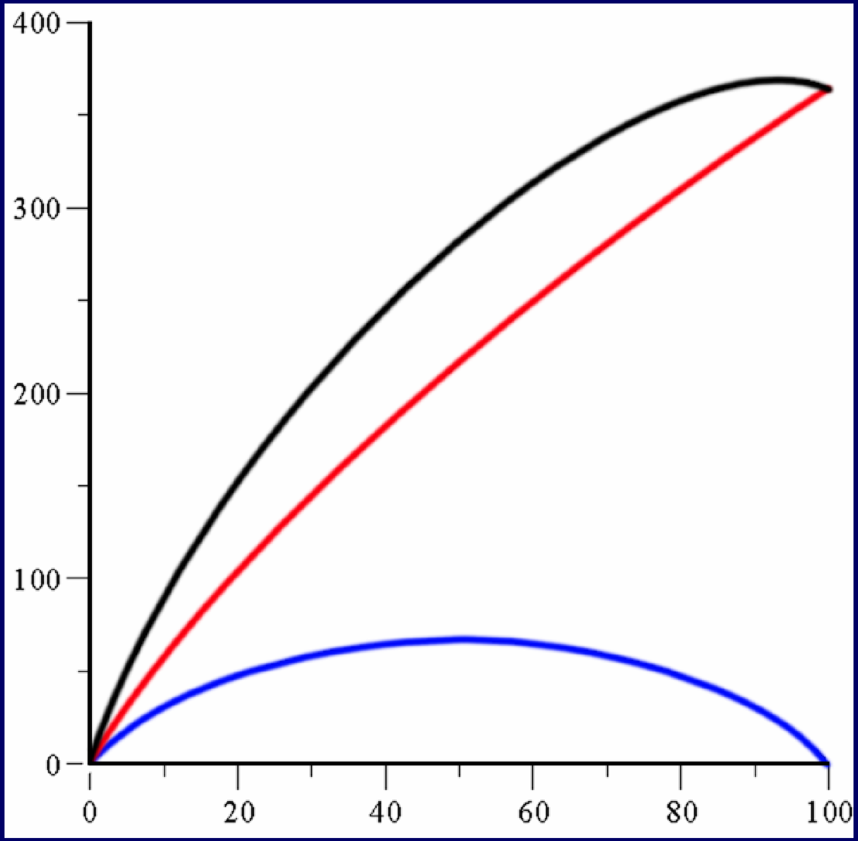
problem 12.9



$$\frac{S}{k_{\mathrm{B}}} = \ln C_{1500}^{n} = \ln \frac{1500!}{(1500-n)!\,n!}$$

$$= \ln \frac{1500 \cdot 1499 \cdot \cdots \cdot (1500-n+1)}{n \cdot (n-1) \cdot \cdots \cdot 1}$$

$$= \ln \frac{1500}{n} + \ln \frac{1499}{n-1} + \cdots + \ln \frac{1500-n+1}{1}$$



$S_{\max}$ at x corresponding equal density, i.e.

$$\frac{x}{1500} = \frac{100 - x}{100} \qquad x = \frac{1500}{16} = 93.75$$

problem 13.2

$$\pm\frac{\Delta\lambda}{\lambda}\approx\beta=\frac{v}{c}\rightarrow\frac{v_{\text{rms}}}{c}$$

room temperature $k_{\mathrm{B}}300\mathrm{K}\approx 1/40\mathrm{eV}$

598nm, **1u***c*²** = 931.494MeV**

$$\frac{v_{\text{rms}}^2}{c^2}=\frac{3k_{\mathrm{B}}T}{mc^2}=\frac{3}{40}\times\frac{1}{23\times 931\times 10^6}$$

$$\Delta\lambda\approx\frac{v_{\text{rms}}}{c}\lambda=0.0011\mathrm{nm}$$

problem 13.6

$$N=\int_{v_\mathrm{m}}^{\infty} n_0\left(\frac{1}{\pi v_\mathrm{m}^2}\right)^{3/2}\exp\left(-\frac{v^2}{v_\mathrm{m}^2}\right)4\pi v^2\,\mathrm{d}v=\frac{4n_0}{\sqrt{\pi}}\int_1^\infty \mathrm{e}^{-x^2}x^2\,\mathrm{d}x$$

$$=\frac{2n_0}{\sqrt{\pi}}\left(\mathrm{e}^{-1}+\int_1^\infty \mathrm{e}^{-x^2}\,\mathrm{d}x\right)=\frac{2n_0}{\sqrt{\pi}}\left(\mathrm{e}^{-1}+\frac{\sqrt{\pi}}{2}-\int_0^1 \mathrm{e}^{-x^2}\,\mathrm{d}x\right)$$

$$=n_0\left(1+\frac{2}{\mathrm{e}\sqrt{\pi}}-\operatorname{erf}(1)\right)$$

$$=n_0\left(1+\frac{2}{\mathrm{e}\sqrt{\pi}}-0.84270\right)=0.57241n_0$$

problem 13.10

$$\overline{E}=\frac{\int \frac{1}{2}mv^2\mathrm{d}\Gamma}{\int \mathrm{d}\Gamma}=\frac{1}{2}m\frac{\int_0^\infty \exp\left(-\frac{v^2}{v_m^2}\right)v^5\mathrm{d}v}{\int_0^\infty \exp\left(-\frac{v^2}{v_m^2}\right)v^3\mathrm{d}v}$$

$$=\frac{1}{2}mv_\mathrm{m}^2\frac{\int_0^\infty \mathrm{e}^{-x^2}x^5\mathrm{d}x}{\int_0^\infty \mathrm{e}^{-x^2}x^3\mathrm{d}x}$$

$$=\frac{1}{2}m\frac{2k_\mathrm{B}T}{m}\frac{\Gamma(3)}{\Gamma(2)}=2k_\mathrm{B}T$$

Alternative

$$\overline{E} = \frac{\int \frac{1}{2} m v^2 \mathrm{d}\mathit{\Gamma}}{\mathit{\Gamma}}$$

$$= \frac{1}{2} m \frac{\boxed{\pi n_0 \left(\frac{m}{2\pi k_{\mathrm{B}} T} \right)^{3/2}} \int_0^\infty \exp\left(-\frac{v^2}{v_m^2} \right) v^5 \mathrm{d}v}{\frac{n_0}{4} \overline{v}}$$

$$\overline{v} = \sqrt{\frac{8 k_{\mathrm{B}} T}{\pi m}}$$

Alternative

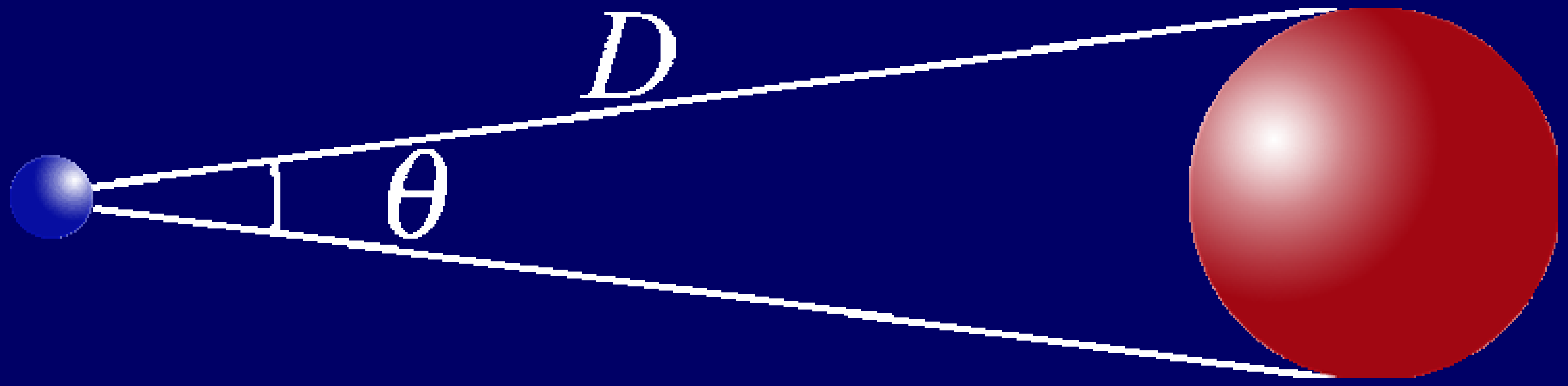
$$\mathrm{d}\Gamma = \pi n_0 \left(\frac{1}{2\pi k_\mathrm{B} T} \right)^{3/2} \mathrm{e}^{-E/k_\mathrm{B}T} \frac{2}{\sqrt{m}} E \mathrm{d}E$$

problem 12.12 from Gibbs-Duheim equation

$$\mathrm{d}\mu = -s\mathrm{d}T + v\mathrm{d}p$$

$$\left(\frac{\partial\mu}{\partial v}\right)_T = \left(\frac{\partial\mu}{\partial p}\right)_T \left(\frac{\partial p}{\partial v}\right)_T = v\left(\frac{\partial p}{\partial v}\right)_T$$

problem 13.12



$$4\pi r^2 \sigma T_b^4 = 4\pi R_\odot^2 \sigma T_\odot^4 \frac{\pi r^2}{4\pi D^2}$$

$$T_{\mathrm{b}} = T_{\odot}\sqrt[4]{\frac{R_{\odot}^2}{4D^2}}$$

$$= T_{\odot}\sqrt{\frac{R_{\odot}}{2D}}$$

$$= T_{\odot}\sqrt{\frac{\theta}{4}}$$

$$= 5700\mathrm{K}\sqrt{\frac{0.50\times\pi}{4\times180}} = 266\mathrm{K}$$

problem 14.2

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

$$\left(\frac{\partial p}{\partial V}\right)_T = -\frac{nRT}{(V - nb)^2} + 2a\frac{n^2}{V^2}\frac{1}{V} = 0$$

$$\left(\frac{\partial^2 p}{\partial V^2}\right)_T = 2\frac{nRT}{(V - nb)^3} - 6a\frac{n^2}{V^2}\frac{1}{V^2} = 0$$

$$\frac{nRT}{(V - nb)^2} = 2a\frac{n^2}{V^2}\frac{1}{V}$$

$$\frac{nRT}{(V - nb)^3} = 3a\frac{n^2}{V^2}\frac{1}{V^2}$$

$$V - nb = \frac{2}{3}V \qquad V_C = 3nb$$

$$\boxed{\begin{aligned} \frac{nRT}{(V-nb)^2} &= 2a\frac{n^2}{V^2}\frac{1}{V} \\ \frac{nRT}{(V-nb)^3} &= 3a\frac{n^2}{V^2}\frac{1}{V^2} \end{aligned}}$$

$$nRT_C = 4n^2b^2 \cdot 2a\frac{n^2}{27n^3b^3}$$

$$RT_C = \frac{8a}{27b}$$

$$p_C = \frac{n\dfrac{8a}{27b}}{2nb} - a\frac{1}{9b^2} = \frac{a}{27b^2}$$

problem 14.3

$$\tilde{p}\boxed{\frac{a}{27b^2}} = \frac{n\tilde{T}\boxed{\frac{8a}{27b}}}{\boxed{3nb}\tilde{V} - nb} - a\frac{n^2}{\boxed{9n^2b^2}\tilde{V}^2}$$

$$\tilde{p} = \frac{8\tilde{T}}{3\tilde{V} - 1} - \frac{3}{\tilde{V}^2}$$

$$\left(\tilde{p} + \frac{3}{\tilde{V}^2}\right)(3\tilde{V} - 1) = 8\tilde{T}$$

Problem 14.5 **Dieterici equation**

$$\ln p + \ln(V - b) = \ln RT - \frac{a}{RTV}$$

$$\frac{1}{p}\left(\frac{\partial p}{\partial V}\right)_T + \frac{1}{V - b} = \frac{a}{RTV^2}$$

At the critical point

$$\left(\frac{\partial p}{\partial V}\right)_T = p\left[-\frac{1}{V-b}+\frac{a}{RTV^2}\right]=0$$

$$-\frac{1}{V-b}+\frac{a}{RTV^2}=0$$

$$\left(\frac{\partial^2 p}{\partial V^2}\right)_T = 0 = p\left\{\frac{1}{(V-b)^2}-\frac{2a}{RTV^3}\right\}$$

$$\frac{1}{(V-b)^2}-\frac{2a}{RTV^3}=0$$

$$2(V-b)=V \quad \text{or} \quad V_C = 2b$$

$$RT_C = \frac{a(V_C - b)}{V_C^2} = \frac{a}{4b}$$

$$\left(\frac{pV}{RT}\right)_C = \left(\frac{V}{V-b}\,e^{-\frac{a}{RTV}}\right)_C = 2e^{-2}$$